HOSPITAL RESOURCE OPTIMIZATION AND DEMAND FORECASTING

Project Id: 25-26J-226

Individual Project Proposal Report

Samadhi Rasanajana Wijethunga

B.Sc. (Hons) Degree in Information Technology
(Specializing in Information Technology)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

June 2025

HOSPITAL RESOURCE OPTIMIZATION AND DEMAND FORECASTING

Project Id: 25-26J-226

Individual Project Proposal Report

B.Sc. (Hons) Degree in Information Technology
(Specializing in Information Technology)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

June 2025

DECLARATION

DECLARATION, COPYRIGHT STATEMENT AND THE STATEMENT OF THE SUPERVISOR.

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Wijethunga S R	IT22179326	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Name of supervisor: Mr. Kanishka Yapa

Signature of supervisor: Date: 27/08/2025

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisors, Mr. Kanishka Yapa and Mr. Uditha Dharmakeerthi, for their valuable guidance and encouragement as we begin this research journey. Their insights and direction have been instrumental in shaping the foundation of our project. I also wish to extend my appreciation to our external supervisor, Ms. Akshi De Silva, for her support and professional advice, which has greatly helped us in setting the right path for this work. Finally, I would like to acknowledge my team members for their collaboration and contribution of ideas. The discussions and shared efforts have already played a vital role in strengthening the early stages of this research.

ABSTRACT

Ensuring the availability of critical medicines at the right time and in the right quantities remains one of the most pressing challenges for hospital systems worldwide. Medicine shortages during emergencies or seasonal illness outbreaks can lead to treatment delays, compromised patient care, and in some cases, preventable deaths. Conversely, overstocking medicines not only strains hospital budgets but also results in wastage through expired drugs and inefficient resource utilization. Traditional inventory management practices in hospitals tend to be reactive and manually driven, responding to shortages only after they occur. This lack of foresight leaves hospitals vulnerable to sudden spikes in demand caused by predictable patterns such as seasonal influenza outbreaks, dengue surges, or public events that increase accident cases.

This research component introduces an AI-powered forecasting and accident surge detection system that leverages both temporal analytics and event-driven insights to predict medicine demand. The system combines historical consumption records with contextual external factors such as festivals, holidays, sporting events, and environmental changes that influence accident rates. By applying time-series models (ARIMA, Prophet, LSTM**)** alongside NLP-based event detection techniques, the framework generates highly contextualized predictions of medicine needs. The resulting forecasts feed into a smart inventory management engine**,** which balances routine requirements with emergency preparedness, ensuring hospitals are both efficient and resilient in the face of demand fluctuations.

A unique feature of this system is the integration of Explainable AI (XAI)**.** While predictive systems have been applied in other industries, their adoption in healthcare often faces barriers of trust and transparency. By using explainability frameworks such as SHAP and LIME, this system provides justifications for its recommendations, making it clear to hospital staff why a demand spike is expected and which medicines should be prioritized. Beyond operational efficiency, the proposed solution contributes to the broader vision of smart hospital management, aligning with global trends in digital healthcare transformation while remaining adaptable to resource-limited contexts such as Sri Lanka.

TABLE OF CONTENTS

DECLARATION i
ACKNOWLEDGEMENT ii
ABSTRACT iii
TABLE OF CONTENTS iv
LIST OF FIGURES v
LIST OF TABLES v
LIST OF ABBREVIATION vi
1. INTRODUCTION 1
1.1. Background & Literature 1
1.2. Research Gap 2
1.3. Research Problem 3
2. OBJECTIVES 4
2.1. Main Objectives 4
2.2. Specific Objectives 5
3. METHODOLOGY 7
3.1. The System Overview Diagram 9
3.1.1. Summary of Technologies 10
3.2. Use Case Diagram 11
3.3. Sequence Diagram 11
3.4. Class Diagram 12
4. REQUIREMENTS 13
4.1. User Requirements 13
4.2. Functional Requirements 13
4.3. System Requirements 14
5. MARKETABILITY 15
6. GANTT CHART 16
7. WORK BREAKDOWN STRUCTURE 16
REFERENCE LIST 17

LIST OF FIGURES

Figure 1 : System Overview Diagram 9
Figure 2 : Use case Diagram 11
Figure 3 : Sequence Diagram 11
Figure 4 : Class Diagram.....12
Figure 5 : Gantt Chart 16
Figure 6 : Work Breakdown Structure.....16

LIST OF TABLES

Table 1 : Abbreviations.....vi
Table 2 : Technologies.....10

LIST OF ABBREVIATION

Table 1 : Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
EHR	Electronic Health Record
HTTPS	Hypertext Transfer Protocol Secure
IoMT	Internet of Medical Things
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
NLP	Natural Language Processing
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
SSL	Secure Sockets Layer
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1. Background & Literature

The demand for medicines within hospitals is not static; rather, it fluctuates based on a combination of seasonal illnesses, public health emergencies, and accident-driven surges. For instance, dengue fever outbreaks typically follow monsoon rains, while influenza cases increase during colder months. Similarly, festive periods often see a rise in firecracker-related injuries, while mass public events like parades, protests, or sports tournaments increase the likelihood of accidents requiring emergency medical care. Hospitals that fail to anticipate these demand patterns often face critical shortages, which directly affect patient safety and disrupt care delivery.

Globally, researchers have begun to explore the application of artificial intelligence and predictive analytics to supply chain and inventory management. Time-series forecasting methods such as ARIMA and Prophet have been widely used in retail and manufacturing to predict demand based on historical trends. More recently, deep learning models such as Long Short-Term Memory (LSTM) networks have shown superior performance in capturing complex temporal dependencies, making them ideal for healthcare contexts where demand patterns exhibit seasonality and irregular spikes. In parallel, natural language processing (NLP) techniques have demonstrated their ability to extract insights from unstructured data sources, such as news reports or public announcements, identifying correlations between external events and demand shifts.

Despite these advancements, the integration of predictive analytics into hospital pharmacy operations remains limited**,** particularly in developing countries. Most hospitals still rely on manual estimates made by pharmacists based on prior experience. While these estimates may work for routine operations, they fall short in anticipating extraordinary demand scenarios, such as mass accident events or simultaneous outbreaks of multiple illnesses. Literature highlights that medicine shortages during emergencies not only compromise patient outcomes but also damage public trust in healthcare systems. Conversely, overstocking leads to significant wastage — the WHO estimates that 10–30% of hospital medicines globally are wasted due to expiry or improper storage.

Therefore, this research aims to bridge the gap between advanced forecasting techniques and practical, resource-efficient hospital operations. By combining temporal demand forecasting with event-driven accident surge detection, it provides hospitals with a proactive tool to optimize inventory while ensuring preparedness for both routine and emergency scenarios.

1.2. Research Gap

While healthcare systems across the globe have explored digital inventory management and basic demand forecasting, several critical limitations continue to hinder effective implementation in hospital pharmacies. In Sri Lanka, these limitations are particularly visible due to the reliance on manual forecasting practices and reactive decision-making. Existing systems primarily focus on replenishing stock after shortages occur, rather than predicting demand in advance. This creates a vicious cycle of emergency procurement, higher costs, and frequent unavailability of essential medicines during seasonal or event-driven surges.

Globally, studies have highlighted those predictive analytics can reduce medicine shortages by more than 25 percent when implemented effectively. However, most of these models are developed for high-resource hospital environments equipped with sophisticated digital infrastructures, such as integrated electronic health record (EHR) systems and advanced pharmacy management software. These conditions do not align with the realities of developing countries like Sri Lanka, where data is often fragmented across multiple sources and maintained in inconsistent formats. Consequently, there is a lack of context-specific models that can operate efficiently with limited resources while still delivering accurate predictions.

Another gap lies in the integration of external event-driven insights. Current forecasting methods in healthcare focus heavily on historical consumption data while ignoring contextual factors such as festivals, sports events, or public gatherings, which have been shown to significantly influence accident rates and, in turn, hospital demand for specific medicines. For example, firecracker-related injuries during the New Year season or traffic-related accidents during long weekends are recurring events that could be forecasted but are often overlooked in traditional systems.

Finally, even where predictive analytics is applied, explainability remains limited. Hospital staff and administrators often hesitate to adopt AI-driven recommendations because they cannot understand the reasoning behind predictions. Without transparent and interpretable explanations, such systems risk being underutilized despite their technical accuracy.

Thus, the research gap can be summarized as follows: while predictive demand forecasting has demonstrated potential in healthcare, there is a clear need for a localized, event-driven, and explainable medicine forecasting system tailored to hospitals in Sri Lanka and similar developing contexts.

1.3. Research Problem

The central research problem addressed in this study revolves around the unpredictability of medicine demand and the inability of hospitals to proactively prepare for sudden surges. Traditional pharmacy systems in Sri Lanka are largely reactive, relying on manual checks and estimates by pharmacists. While these methods work under stable conditions, they fail during extraordinary events such as seasonal outbreaks or public accidents. This leads to life-threatening shortages, increased patient waiting times, and compromised treatment outcomes.

More specifically, hospitals lack the ability to integrate multi-source data into their forecasting strategies. Medicine usage logs, seasonal disease records, event calendars, and accident intake data are available in some form but are rarely connected to generate predictive insights. As a result, hospitals continue to operate in a siloed fashion, where pharmacy management, emergency response, and hospital operations are disconnected.

The complexity of this research problem lies in its multidimensional nature. It is not sufficient to forecast medicine demand based only on historical patterns, as this ignores the critical role of contextual triggers such as public holidays or environmental changes. At the same time, the solution must be interpretable and usable by non-technical staff, ensuring that predictions translate into actionable stocking strategies.

Thus, the research problem can be articulated as:
"How can hospitals develop an AI-driven system that combines temporal forecasting, NLP-based event detection, and explainable AI to proactively manage medicine demand and accident surges, ensuring preparedness and reducing both shortages and wastage?"

2. OBJECTIVES

2.1. Main Objectives

The primary objective of this research component is to design and implement a predictive analytics framework that enhances the efficiency and reliability of hospital pharmacy operations by forecasting medicine demand and detecting accident-driven surges. In many healthcare institutions, pharmacy inventory is managed through static ordering schedules or reactive adjustments made after shortages occur. These approaches leave hospitals vulnerable to crises, particularly during seasonal outbreaks or when sudden accidents generate unexpected patient inflows.

This research aims to address these challenges by introducing an intelligent, data-driven solution that integrates historical consumption data, accident intake records, seasonal illness trends, and external event signals into a unified forecasting model. By leveraging advanced time-series forecasting methods such as ARIMA, Prophet, and LSTM, alongside natural language processing (NLP) techniques for event detection, the system will generate contextualized predictions of medicine demand. These forecasts will then be integrated into a smart inventory optimization engine, ensuring hospitals are able to balance routine needs with surge preparedness.

A key innovation in this objective is the integration of explainable AI. While AI systems are powerful, they are often treated as “black boxes” in healthcare, which limits trust and adoption. By embedding interpretability frameworks such as SHAP and LIME, the proposed system will provide transparent reasoning behind forecasts — for example, explaining that an increase in painkiller demand is predicted because of an upcoming sports event combined with historical accident data. This transparency will help hospital staff make informed decisions and foster greater trust in the system.

Ultimately, this research component contributes not only to operational efficiency but also to the broader goals of patient safety, healthcare preparedness, and the creation of resilient hospital systems capable of anticipating both routine and emergency medicine needs.

2.2. Specific Objectives

To achieve the main objective, six specific objectives have been identified:

1. **Data Collection and Preprocessing**
    - Collect historical hospital pharmacy data, including medicine usage logs, inventory records, seasonal disease reports, and accident intake data.
    - Gather external event data such as festivals, public holidays, sporting events, and political rallies from online calendars, APIs, and news feeds.
    - Standardize medicine data by aligning categories, dosage units, and time stamps across different sources. This ensures that forecasting models can process data consistently and accurately.

2. **Temporal Forecasting Model Development**
    - Design and train time-series models such as ARIMA, Prophet, and LSTM to capture daily, weekly, and seasonal medicine consumption patterns.
    - Incorporate both short-term and long-term consumption trends into the predictions, allowing hospitals to plan for immediate needs while also anticipating seasonal surges.
    - Validate the models using accuracy metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and cross-validation.

3. **Accident Surge Detection through NLP**
    - Implement an NLP pipeline to analyze unstructured text from news reports and event announcements.
    - Extract accident-related keywords such as “parade,” “fireworks,” or “motor rally” and classify them into relevant categories.
    - Develop classification models such as decision trees or logistic regression to map identified events to potential medicine requirements, for example linking fireworks to burn ointments and antibiotics, or sporting events to trauma medicines.

4. **Smart Inventory Optimization**
    - Develop an optimization engine that combines predicted demand with current stock levels to recommend reorder points, safety stock thresholds, and emergency procurement strategies.
    - Ensure that the system balances between routine supply and surge readiness, avoiding both shortages and overstocking.
    - Include additional factors such as cost-effectiveness and expiry dates to ensure efficient resource utilization.

5. **Integration of Explainable AI**
    - Apply explainability frameworks such as SHAP and LIME to make predictions interpretable.
    - Provide hospital staff with clear explanations of why demand spikes are forecasted and which medicines are prioritized.
    - Build trust and confidence among pharmacists and administrators by ensuring that the system is not only accurate but also transparent.

6. **System Evaluation and Validation**
    - Conduct a pilot implementation in a large-scale hospital environment.
    - Compare predicted demand against actual medicine usage to assess accuracy and reliability.
    - Evaluate impact using operational metrics such as reduction in stockouts, decrease in wastage, and improved emergency response.

Gather feedback from pharmacists, inventory managers, and hospital administrators to refine usability and ensure the system integrates smoothly into hospital workflow.

3. METHODOLOGY

The methodology adopted in this research follows a structured, multi-stage approach that ensures systematic development, testing, and validation of the forecasting and surge detection system. Each stage contributes to building a robust and context-specific solution tailored to hospital pharmacy operations.

Stage 1: Problem Understanding and Data Collection

The first stage involves identifying the specific challenges faced by hospital pharmacies in managing inventory. Through interviews with pharmacists, hospital administrators, and emergency care staff, the key issues of shortages, overstocking, and delayed restocking will be analyzed. Multi-source data will then be collected, including historical medicine consumption logs, pharmacy inventory records, seasonal illness data, accident and emergency intake reports, and external event information. Public calendars, government announcements, and news feeds will be used to capture information about upcoming festivals, holidays, and events that may influence medicine demand.

Stage 2: Data Preprocessing

The raw data collected is often fragmented and inconsistent, making preprocessing essential. Data cleaning will involve handling missing values, correcting inconsistencies, and standardizing medicine names, dosage units, and categories. Event data will be aligned with hospital timelines to ensure consistency in forecasting. For unstructured data such as news articles, NLP techniques will be used to extract relevant keywords (e.g., “traffic accident,” “festival,” “fireworks”) and classify them into categories associated with accident-related surges.

Stage 3: Feature Engineering

To enhance model accuracy, feature engineering will incorporate both temporal and contextual indicators. Time-based features such as day of the week, month, and season will be added, alongside event-based flags marking festivals, public holidays, or known accident-prone periods. Public health indicators, such as local outbreak notices or climate conditions, will also be included. Lag features and moving averages will be used to capture short-term trends in medicine usage.

Stage 4: Predictive Modeling

Time-series models such as ARIMA and Prophet will be used to generate baseline forecasts of medicine demand. LSTM models will then be applied to capture complex non-linear relationships and long-term dependencies. For event-driven predictions, classification models such as decision trees and logistic regression will be used to map detected events to specific medicine categories. These models will be evaluated using accuracy metrics and cross-validation to ensure robustness. The final forecasting system will integrate both temporal and event-driven predictions into a unified framework.

Stage 5: Accident Surge Detection

The system will match detected events with historical accident patterns to predict accident-related surges. For example, firecracker festivals will be associated with increased demand for burn ointments and antibiotics, while rainy seasons combined with sporting events may signal an increase in trauma-related medicine needs. This integration of event-based insights with temporal forecasting represents a key innovation of the system.

Stage 6: Smart Inventory Optimization

Based on the forecasts, a smart inventory optimization engine will be developed. This engine will calculate reorder points, safety stock levels, and procurement recommendations. It will generate automated alerts when shortages are predicted, highlight urgent restocking needs, and balance routine requirements with emergency preparedness. The system will also incorporate cost and expiry considerations, ensuring medicines are used efficiently and wastage is minimized.

Stage 7: Explainable AI Integration

To address the issue of transparency, explainable AI frameworks such as SHAP and LIME will be embedded into the system. These frameworks will provide clear justifications for predictions, showing hospital staff which features (e.g., event type, seasonal illness trends) contributed to a particular forecast. For example, the system may explain that “demand for antibiotics is predicted to rise due to the upcoming rainy season combined with historical dengue outbreak data.”

Stage 8: Evaluation and Validation

The system will be pilot tested in a large-scale hospital environment to assess its performance in real-world conditions. Predictions will be compared against actual medicine usage data, and accuracy metrics such as RMSE and MAE will be calculated. Operational metrics, such as reduction in stockouts, decreased wastage, and improved emergency readiness, will also be evaluated. Feedback from pharmacists and administrators will be gathered to refine system usability and integration into hospital workflows.

3.1. The System Overview Diagram

The proposed system follows a modular architecture with four main layers. First, data collection and preprocessing layer integrates structured data such as medicine usage logs and inventory records with unstructured data from event calendars and news sources, cleaning and standardizing them for analysis. Next, the predictive modeling layer uses ARIMA, Prophet, and LSTM models for temporal forecasting, while an NLP-based module detects event-driven surges; their outputs are combined for accurate demand prediction. A smart inventory optimization engine then translates forecasts into actionable strategies such as reorder points, safety stock levels, and surge alerts. Finally, an Explainable AI layer, using SHAP and LIME, provides transparent reasoning for predictions, displayed through a user-friendly dashboard for hospital staff.

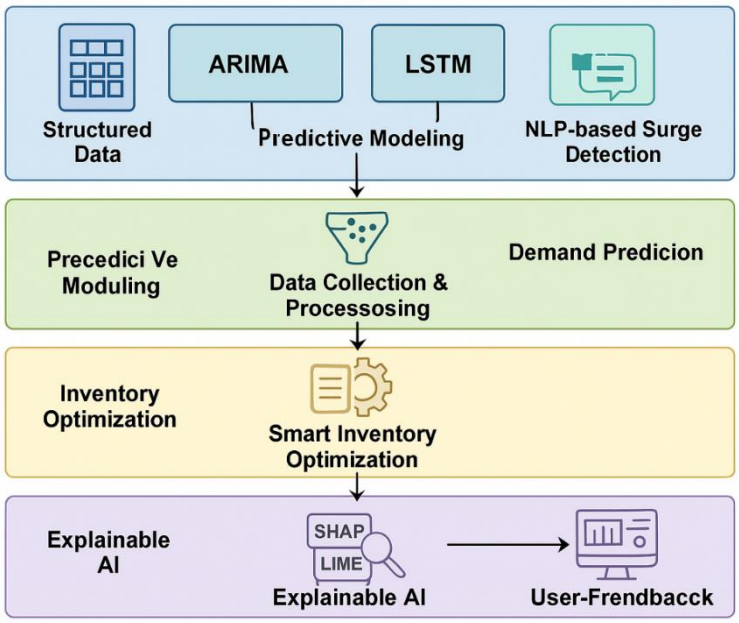

Figure 1 : System Overview Diagram

3.1.1. Summary of Technologies

Summary of technologies, techniques, architectures and algorithms used is shown in the table below.

Table 2 : Technologies

Description	Tools and Technologies
Frontend Technologies	React Native
Data Analysis	Python (Pandas, NumPy)
Backend Technologies	NodeJS, Python
Machine learning	scikit-learn, TensorFlow
Database for store data	MongoDB
Hosting the API	Heroku
Weather API	OpenWeatherMap API
Version Controlling	Git
Team connectivity	Teams and WhatsApp

3.2. Use Case Diagram

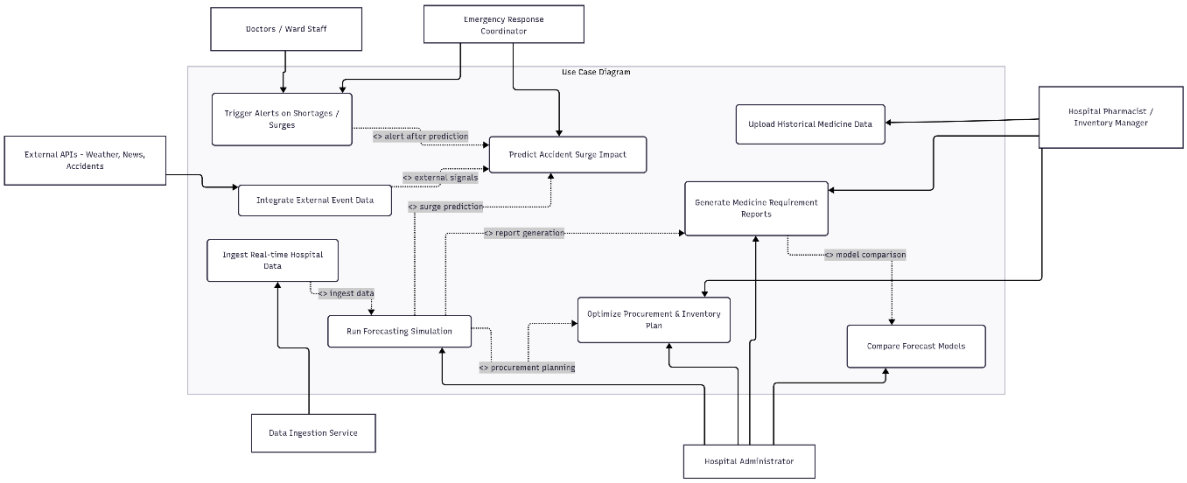

Figure 2 : Use case Diagram

3.3. Sequence Diagram

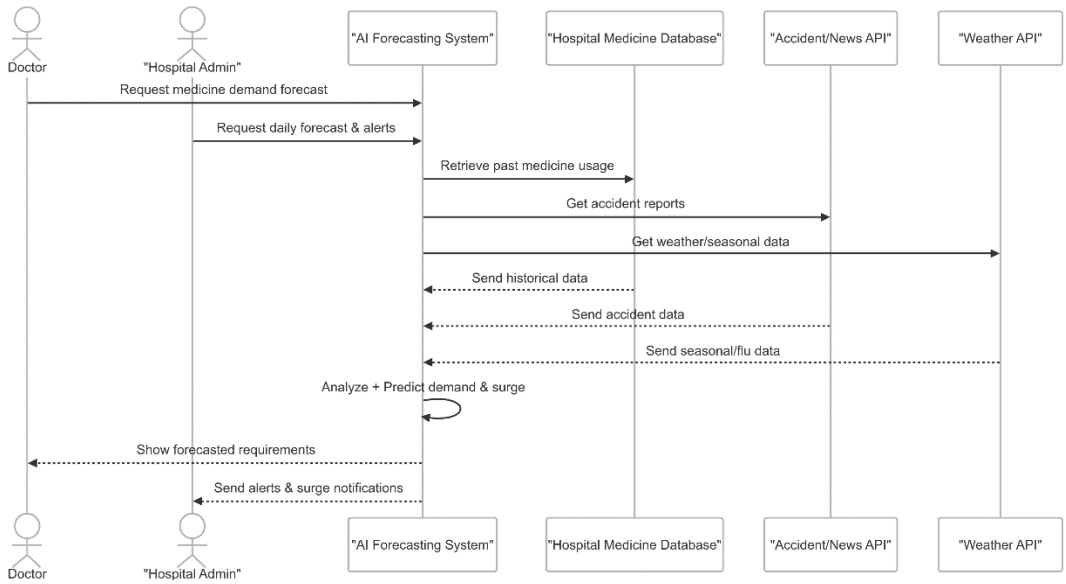

Figure 3 : Sequence Diagram

3.4. Class Diagram

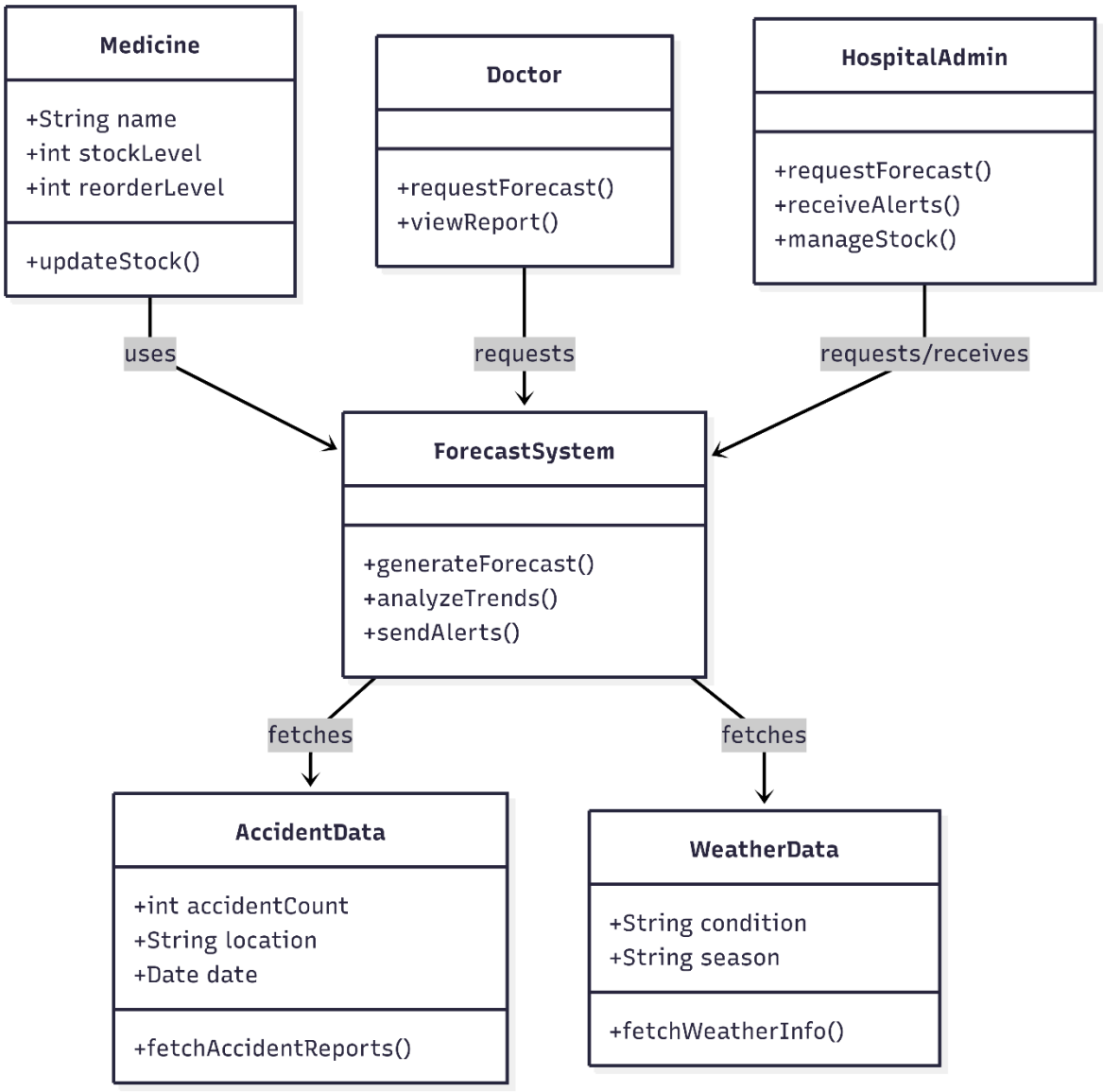

Figure 4 : Class Diagram

4. REQUIREMENTS

The requirements for this system are categorized into user, functional, and system requirements. These requirements ensure that the system addresses the needs of hospital staff while being technically feasible and sustainable.

4.1. User Requirements

From the perspective of hospital pharmacists, inventory managers, and administrators, the system must provide a user-friendly interface that translates complex AI predictions into actionable decisions. Users require the system to:

1. Collect and integrate data from multiple sources, including hospital logs and external event feeds.
2. Forecast medicine demand across short-term (daily/weekly) and long-term (monthly/seasonal) horizons.
3. Detect upcoming events that may trigger accident surges and map them to medicine categories.
4. Generate stocking recommendations that are easy to understand and implement.
5. Provide alerts and notifications for potential shortages in advance.
6. Ensure transparency by explaining why a forecast was made.
7. Be accessible via a web-based dashboard, compatible with desktops, tablets, and mobile devices.

4.2. Functional Requirements

The system must deliver several core functions:

- Performance: The system should generate forecasts and inventory recommendations within seconds, supporting real-time decision-making. It should also support multiple concurrent users without degrading performance.
- Accuracy: Forecasts must reflect actual medicine demand as closely as possible. The event detection module should correctly identify accident-prone events and link them to relevant medicine needs.
- Reliability: The system should operate consistently, with robust mechanisms for handling errors and recovering from failures without data loss.
- Scalability: It must be able to handle increasing data volumes as hospitals collect more detailed records and expand the system to additional wards or hospitals.

4.3. System Requirements

1. Hardware Requirements
   - Minimum and recommended specifications for servers or computers running the system.
   - Sufficient storage capacity for historical data, model parameters, and scheduling logs.
2. Software Requirements
   - Compatible operating systems (e.g., Windows, Linux).
   - Required libraries and frameworks (e.g., TensorFlow, PyTorch, scikit-learn).
3. Database Requirements
   - A robust database system (e.g., PostgreSQL, MongoDB) for storing raw data, processed features, and model outputs.
   - Scalable storage for growing datasets and model artifacts.
4. Network and Connectivity
   - Reliable internet connectivity for accessing external data sources and updates.
   - Network infrastructure supports real-time data exchange and multi-user access.
5. User Interface Requirements
   - A user-friendly interface for hospital staff to interact with predictions and scheduling tools.
   - Compatibility with major web browsers and devices (desktop, tablet, mobile).
6. Security Requirements
   - Secure data transmission using encryption protocols (e.g., HTTPS, SSL).
   - Role-based access control with user authentication and authorization.
7. Performance Requirements
   - Fast response times for generating predictions and scheduling suggestions.
   - Capacity to handle a defined number of concurrent users or requests.

8. Scalability Requirements

- Ability to scale with increasing data volume and user demand.

Support for cloud-based deployment to enhance flexibility and scalability.

5. MARKETABILITY

The proposed Forecast Medicine Requirements and Accident Surge Detection System, has strong potential in the healthcare market because it solves real problem hospitals face every day. Right now, most hospitals in Sri Lanka and even many abroad still manage medicine supplies in a very reactive way. They usually wait until stocks run low or an emergency happens, and only then start the procurement process. This often leads to shortages during dengue seasons, flu outbreaks, or even during sudden accident surges.

What makes this system marketable is that it takes a proactive approach. By using historical data, real-time hospital inputs, and even signals from external events like weather or accidents, it can predict future medicine needs in advance. This means hospitals won't get caught off guard.

The target users are both public and private hospitals, as well as government health departments, since they are under constant pressure to make sure patients always get timely treatment. The system is also flexible—it can work for big hospitals in cities with advanced IT systems, and it can be scaled down for smaller rural hospitals too.

Another strength is trust. Because the system uses explainable forecasting and not just a "black box" AI, doctors and pharmacists can clearly see why certain predictions are made. Plus, it integrates smoothly with existing hospital systems, which makes adoption easier.

In the long run, this could even become a **national platform** for medicine demand forecasting, helping the government and hospitals coordinate better during seasonal outbreaks or emergencies. So, it's not just a hospital product, it can actually become a public health game-changer.

6. GANTT CHART

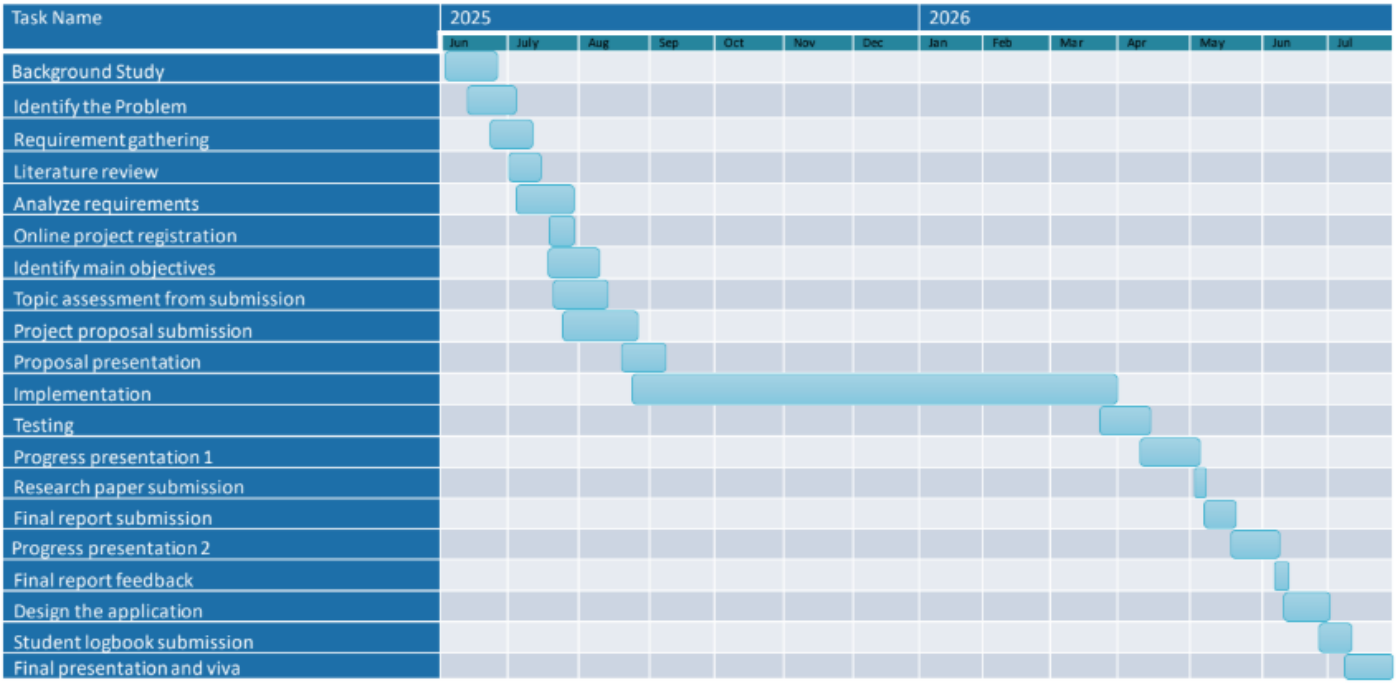

Figure 5 : Gantt Chart

7. WORK BREAKDOWN STRUCTURE

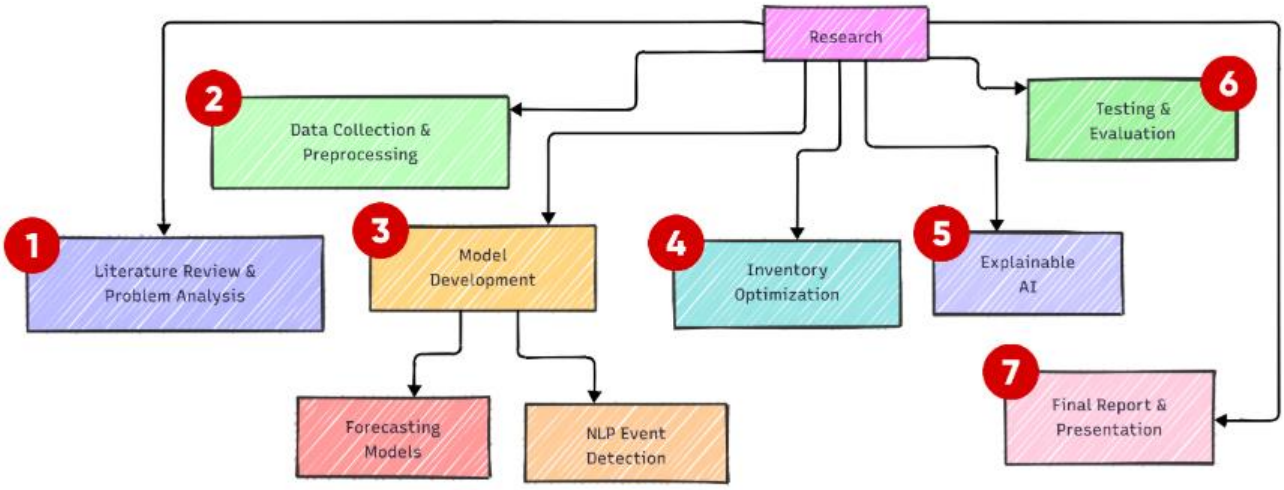

Figure 6 : Work Breakdown Structure

REFERENCE LIST

[1] A. Khan, P. Sharma, and Y. Li, "Explainable AI for hospital resource planning: Improving trust in predictive healthcare systems," *Artificial Intelligence in Medicine*, vol. 154, p. 102358, Mar. 2025.

[2] S. Ghosh, R. Banerjee, A. Das, and S. K. Bandyopadhyay, "Time series forecasting of hospital medicine demand during epidemic outbreaks using ARIMA and Prophet models," *Healthcare Analytics*, vol. 5, pp. 100–112, 2025.

[3] M. Uddin, F. Alam, and T. Ghosh, "AI-driven healthcare supply chain optimization: Forecasting medical resources under demand surges," *IEEE Access*, vol. 13, pp. 15432–15445, 2025.

[4] L. Zhang, J. Chen, and H. Li, "Event-driven healthcare analytics: Using NLP to predict medical demand from public event data," *Journal of Biomedical Informatics*, vol. 150, p. 104612, Jan. 2025.

[5] K. Yapa, A. Jayakody, S. Wijayasekara, S. Kerrison, and D. P. P. Rathu Baduge, "AI and SDN Based Framework to Mitigate Threats and Vulnerabilities in Internet of Medical Things," *International Journal on Communications Antenna and Propagation (I.RE.C.A.P.)*, vol. 15, no. 2, pp. 112–121, Apr. 2025. doi: 10.15866/irecap.v15i2.25994.

[6] World Health Organization, "Medical supply chain and resource preparedness during emergencies," *WHO Technical Report Series*, Geneva, Switzerland, 2024.